Unit 1 Classwork — Real Numbers

Converting · Classification · Squares · Estimating Roots · Ordering

Name _____ Date _____ Period _____

A. Converting Fractions ↔ Decimals

Problem 1

Convert $\frac{3}{8}$ to a decimal.

answer _____

Problem 2

Convert $\frac{5}{6}$ to a decimal (round to the thousandths, use a repeating bar notation if needed).

answer _____

Problem 3

Convert 0.45 to a fraction in simplest form.

answer _____

Problem 4

Convert $\frac{7}{20}$ to a decimal.

answer _____

Problem 5

Convert $0.\overline{6}$ (repeating) to a fraction.

answer _____

Problem 6

Convert $\frac{1}{3}$ to a decimal.

answer _____

B. Classifying Numbers

Problem 1

Is √16 rational or irrational?

answer _____

Problem 2
Is √7 rational or irrational?

answer

Problem 3
Is 0.454545... (repeating) rational or irrational?

answer

Problem 4
Is π rational or irrational?

answer

Problem 5
Is −5 a whole number, an integer, or both?

answer

Problem 6
Is √36 rational or irrational?

answer

C. Perfect Squares

Problem 1
What is 9^2?

answer

Problem 2
What is 12^2?

answer

Problem 3
What is √121?

answer

Problem 4
What is √225?

answer

D. Estimating Square Roots

Problem 1

Between which two consecutive integers does $\sqrt{50}$ fall?

answer

Problem 2

Estimate $\sqrt{50}$ to the nearest tenth.

answer

Problem 3

Between which two consecutive integers does $\sqrt{90}$ fall?

answer

E. Ordering Rational & Irrational Numbers

Problem 1

Order from least to greatest: $\sqrt{10}$, 3, $\frac{10}{3}$

answer

Problem 2

Which is greater: $\sqrt{2}$ or $\frac{3}{2}$?

answer

Problem 3

Which is greater: π or $\sqrt{10}$?

answer